HuBMAP in the CFDE: Discovering, Visualizing, and Reusing Multimodal Omics Data

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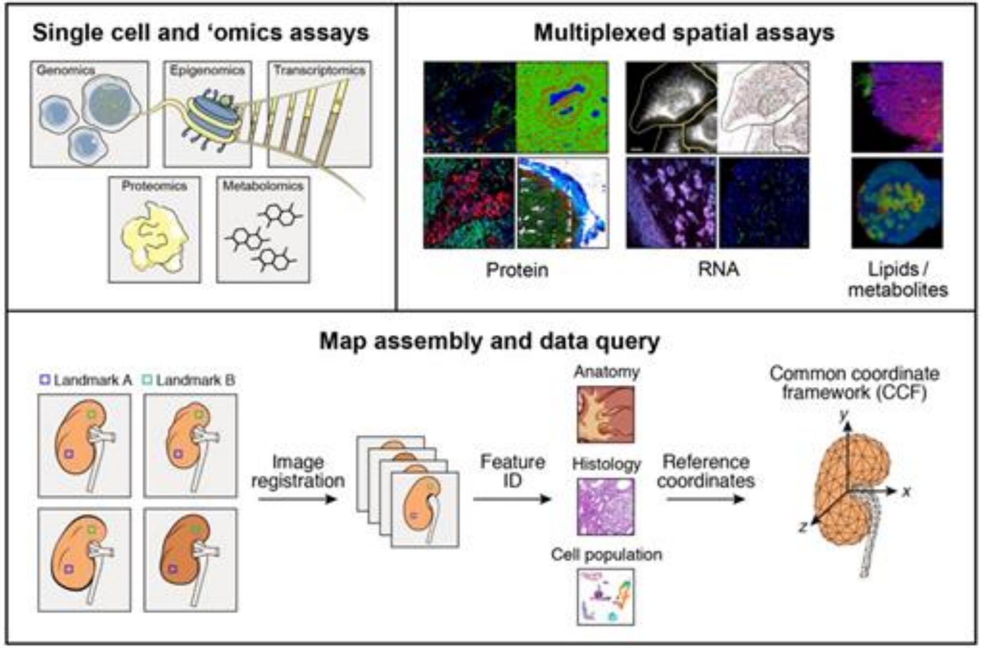
"While we broadly know how cells are organized in most tissues, a comprehensive understanding of the cellular and molecular states and interactive networks resident in the tissues and organs [...] is lacking."

"The specific three-dimensional organization of different cell types, together with the effects of cell–cell and cell–matrix interactions in their natural milieu, have a profound impact on normal function, natural ageing, tissue remodelling, and disease progression in different tissues and organs."

"HuBMAP is an important part of the international mission [and] aspires to help to build a foundation by generating a high-resolution atlas of key organs [...] as well as acting as a key resource for new contributions in the growing fields of tissue biology and cellular ecosystems."



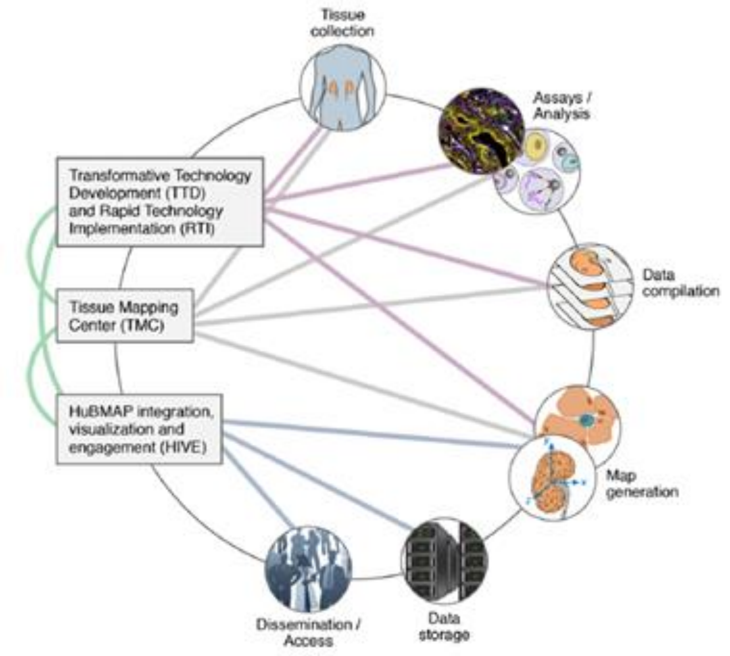
Snyder et al. Nature. 2019; 574(7777)

HuBMAP Vision and Goals

Goal: Catalyze the development of a framework for mapping the human body at single cell resolution

By:

1. Accelerating development of next generation tools and techniques
2. Generating foundational 3D human tissue maps
3. Establishing an open data platform
4. Collaborating with the research community
5. Supporting pilot projects that demonstrate value of HuBMAP resources



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The HuBMAP Consortium

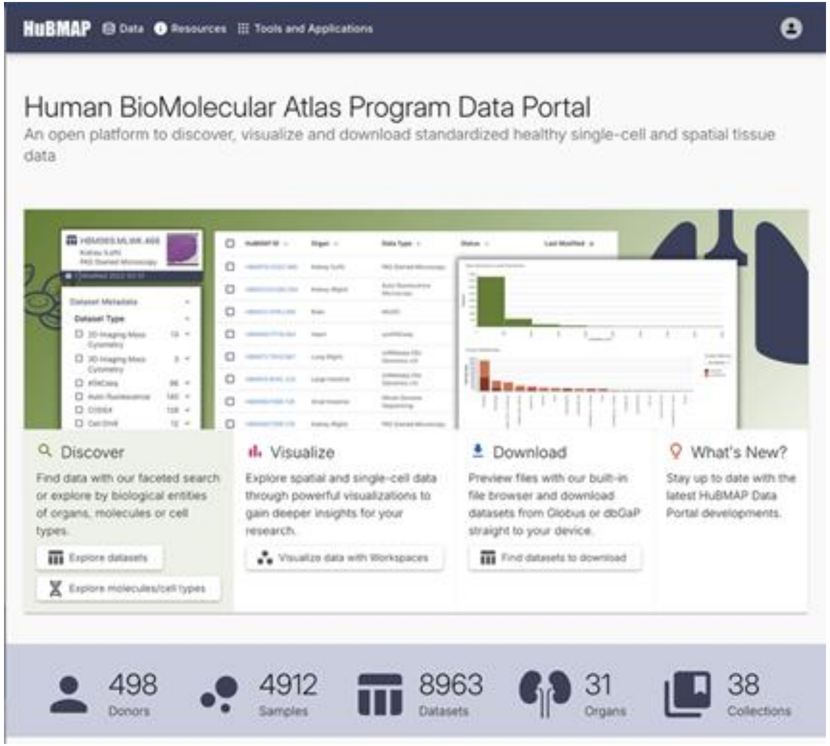
Key Features

- Free, **open access** to non-identifiable data, tools and technologies developed
- **Healthy** tissue
  - Diversity in samples: sex, ethnicity, age
- Expansive, accessible **three-dimensional** molecular and cellular atlas of the human body
- Culture of **openness and sharing** using **team science-based** approaches
  - Protocols.io, github
- **Diverse expertise in funded members**
  - molecular, cellular, developmental, and computational biologists, measurement experts, clinicians, pathologists, anatomists, biomedical and software engineers, and computer and data information scientists

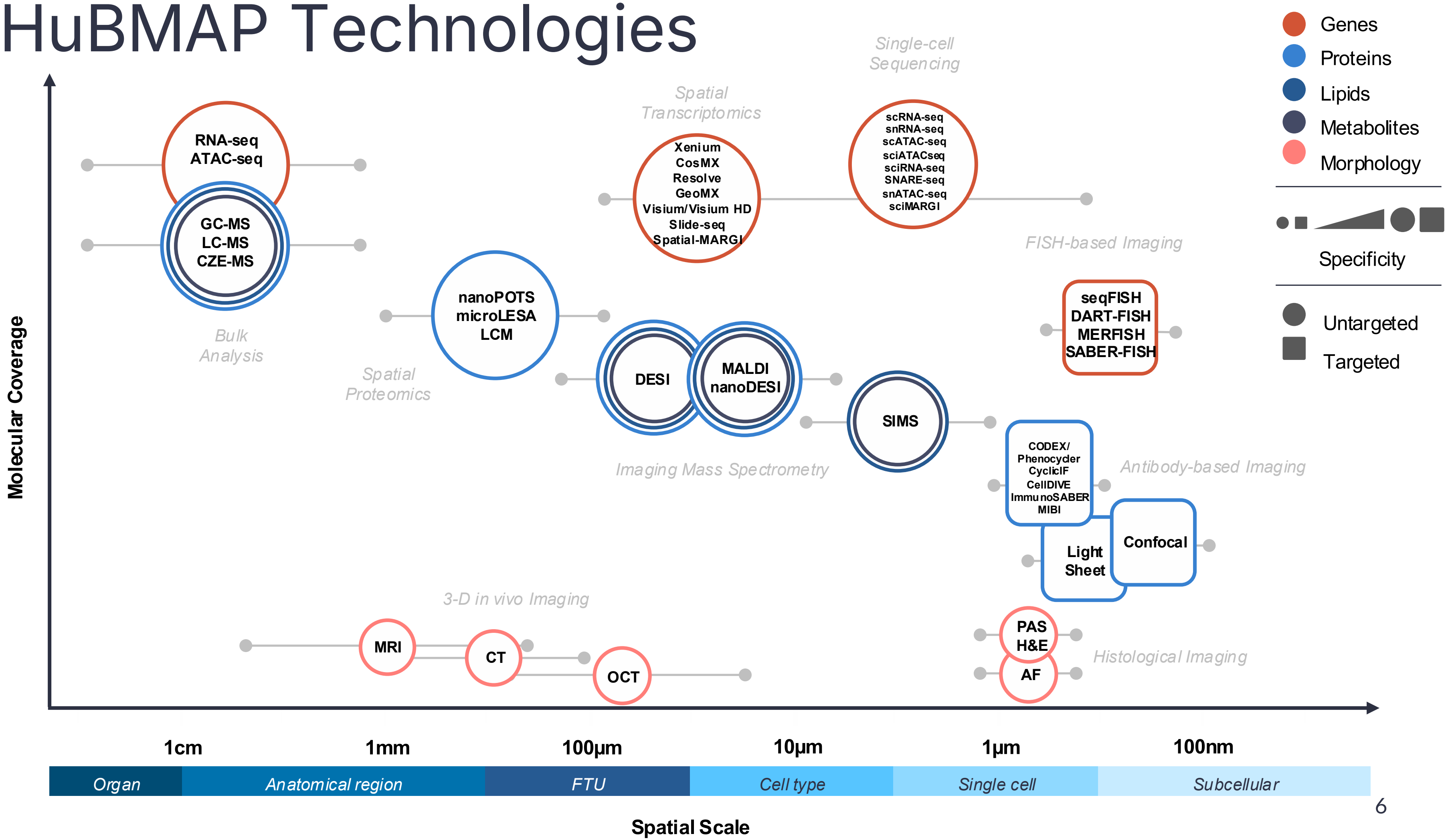
- Coupled with no-cost access leading **HPC resources** and cloud burstability.

The HuBMAP Data Portal: https://portal.hubmapconsortium.org/

Serves as a **repository** for multi-modal spatial and single-cell data from **healthy human tissues** to generate a high-resolution atlas of the human body

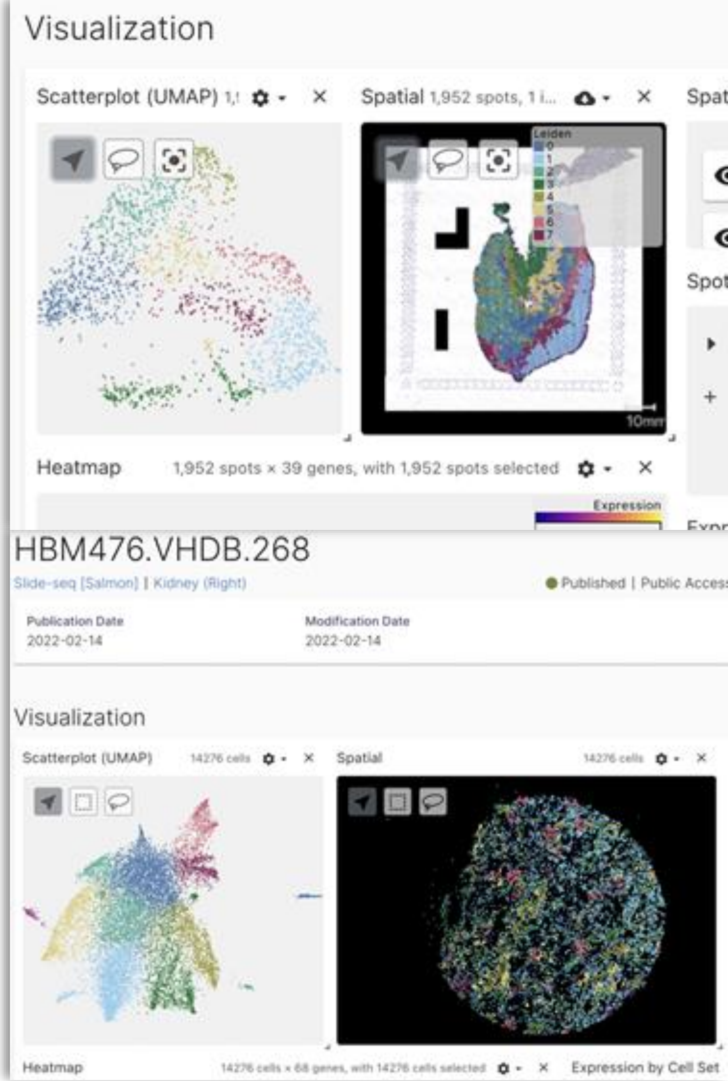


HuBMAP Technologies



HuBMAP Analysis Pipelines

- HuBMAP analysis pipelines support multiple assay families:
  - Single-cell / single-nucleus sequencing
    - sc/snRNA-seq
    - sc/snATAC-seq
    - multiome RNA + ATAC
  - Spatial transcriptomics
    - Visium
    - Slide-seq
  - Imaging and spatial proteomics
    - PhenoCycler
    - CODEX
    - CellDIVE
    - MIBI
    - FISH-based assays
- These pipelines help produce standardized, reusable outputs for Portal discovery, visualization, and downstream analysis.

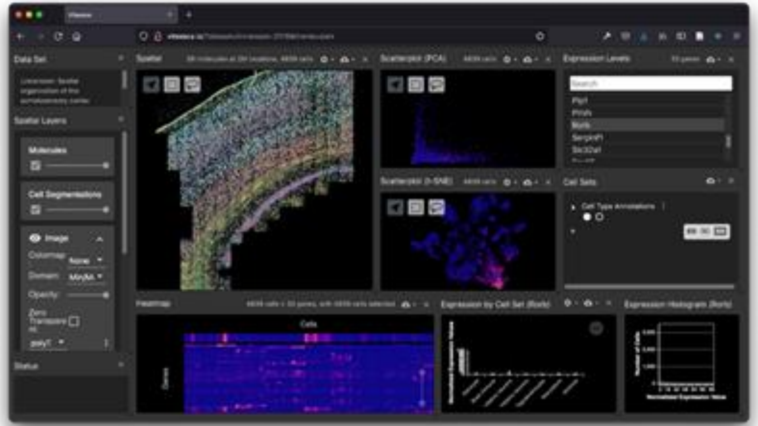


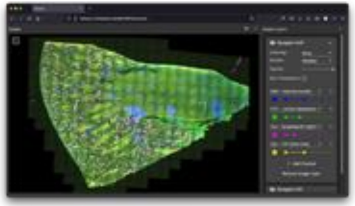
Vitessce

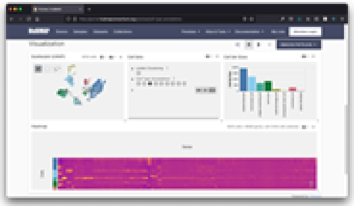
Visual integration tool for exploration of spatial single cell experiments

https://vitessce.io/

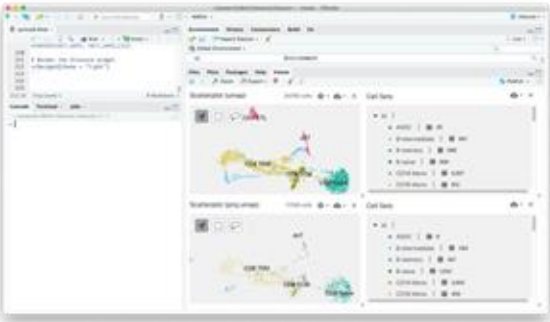
- Modular interactive visualization framework for single-cell data
- Lack of specialized server infrastructure
- Coordination of common entities across views
- Support for data analysis environments (Python/R)
- Support for common single-cell data structures/file formats







Web browser



Keller, M.S., Gold, I., McCallum, C. *et al*. Vitessce: integrative visualization of multimodal and spatially resolved single-cell data. *Nat Methods* 22, 63–67 (2025). https://doi.org/10.1038/s41592-024-02436-x

HuBMAP Omics Data: Discovery and Reuse

Key Portal Requirements

Meet needs of a breadth of users

Portal user interfaces must serve a wide range of user audiences, needs, and workflow entry points particularly for features related to:

- **Search** and **Discovery** of data
- **Visualization** and **Analysis** of data directly in the portal
- **Download** of large collections of data

Portal tools and **infrastructure** must:

- **Support data** from multiple individuals at a range of anatomical scales (organ to subcellular)
- Follow findable, accessible, interoperable, and reusable **(FAIR) data principles**
- Maintain accuracy and **reproducibility**; protect data with appropriate levels of **privacy**
- Follow **community-developed data standards** for algorithms to run at scale
- Ensure **long-term availability** of open tools, data, and infrastructure

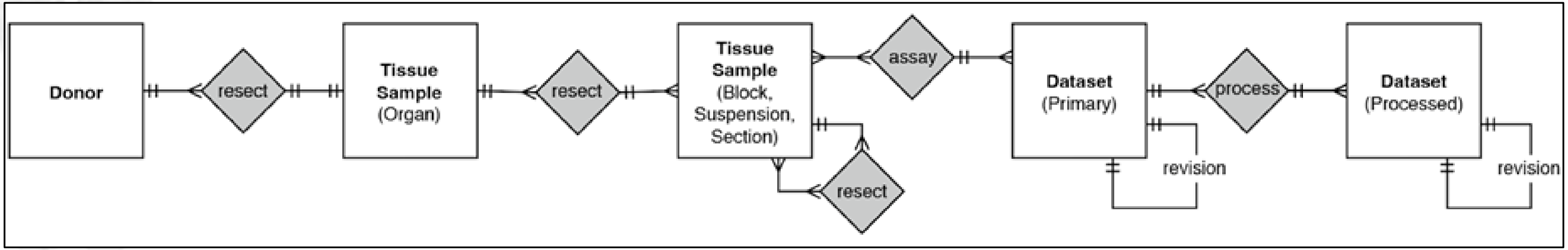
Data on the HuBMAP Data Portal

Biological entities

Donor > organ> sample > datasets (raw and processed)

Provenance is represented on the portal

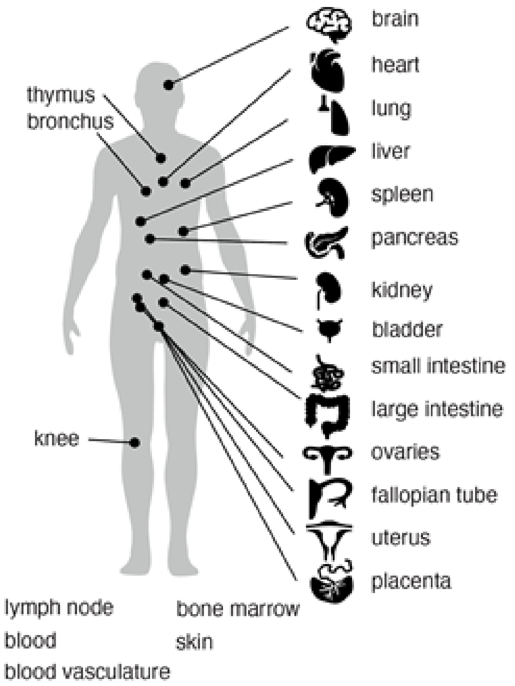
- Tracks entire lifecycle of biological samples and data
- Where data originated, how they've been processed, all associated metadata

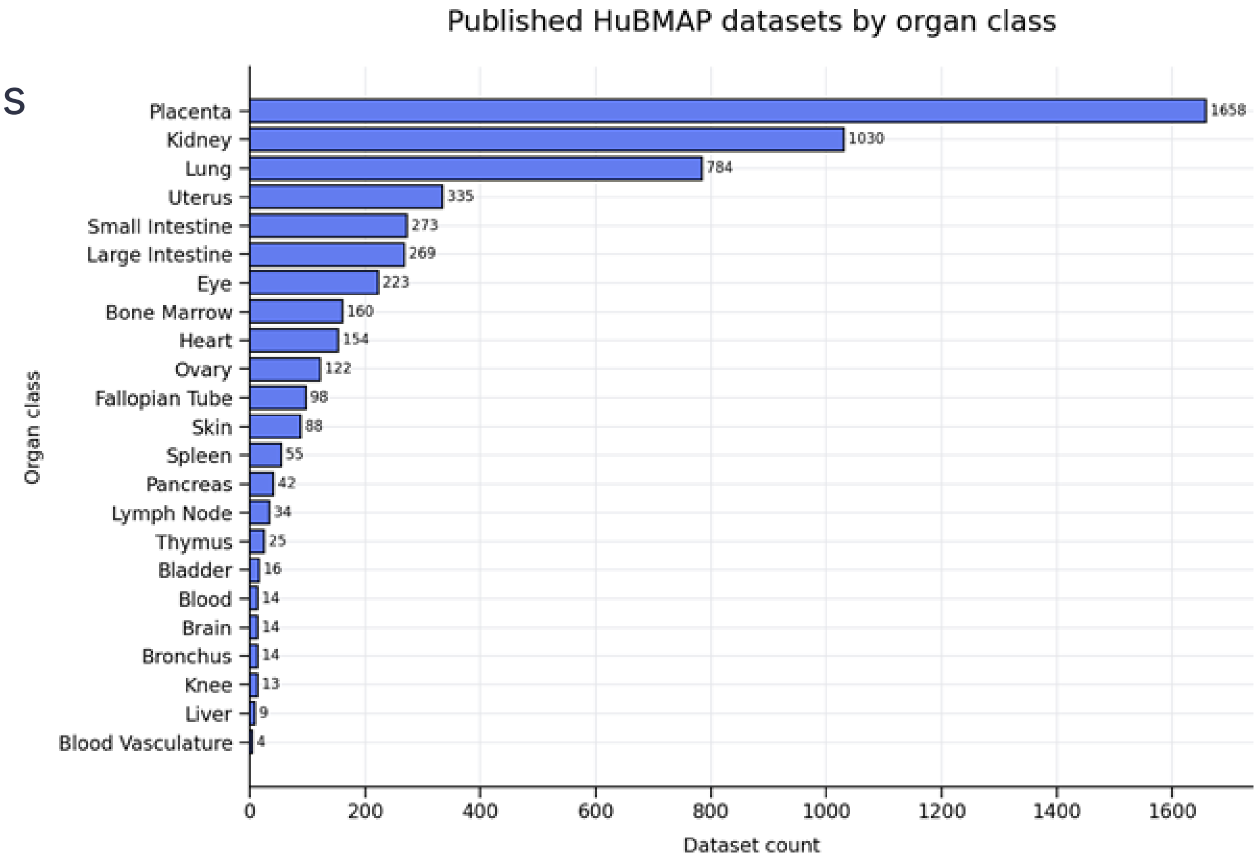


Organs

28 organ classes

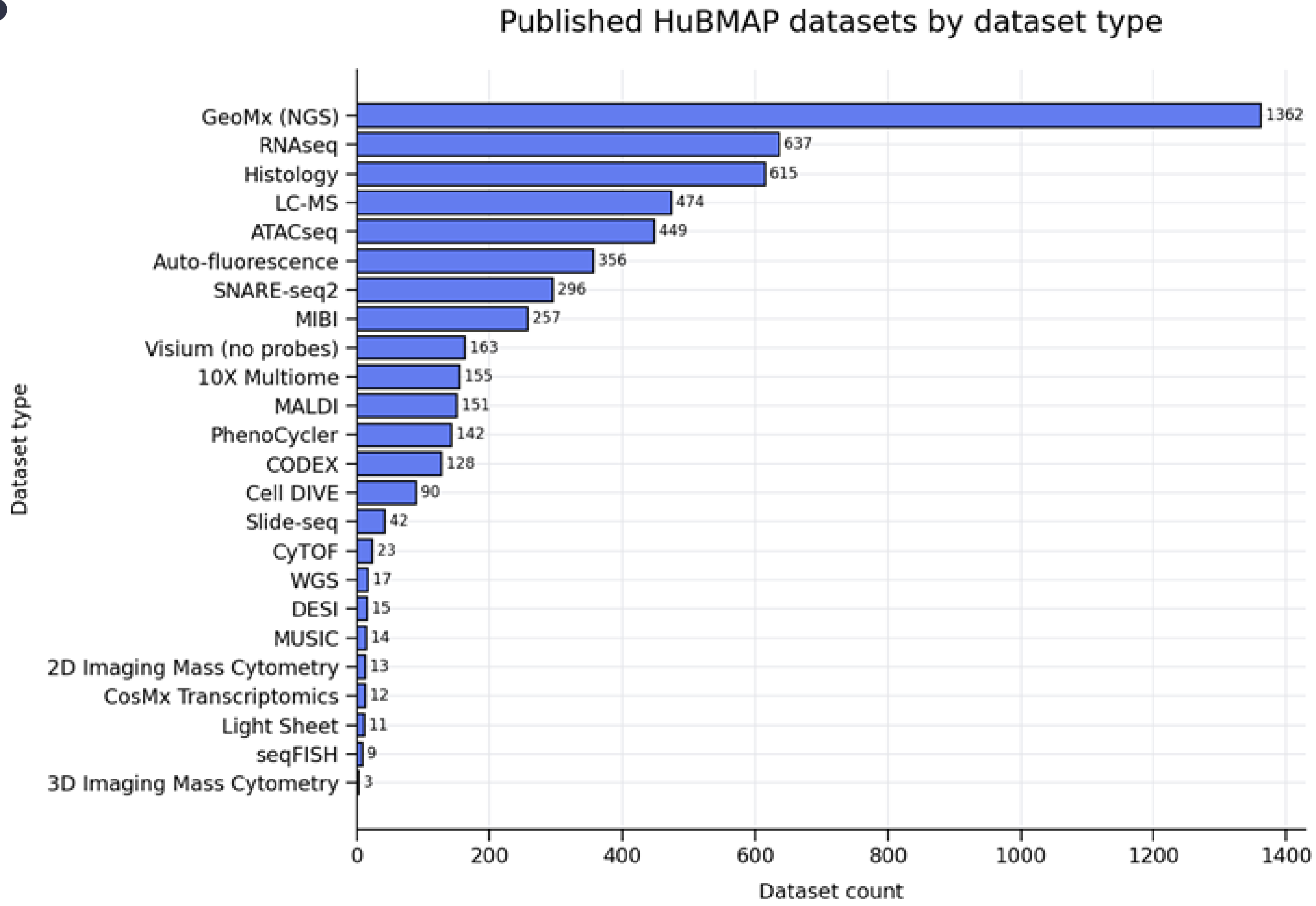
400+ unique donor organs



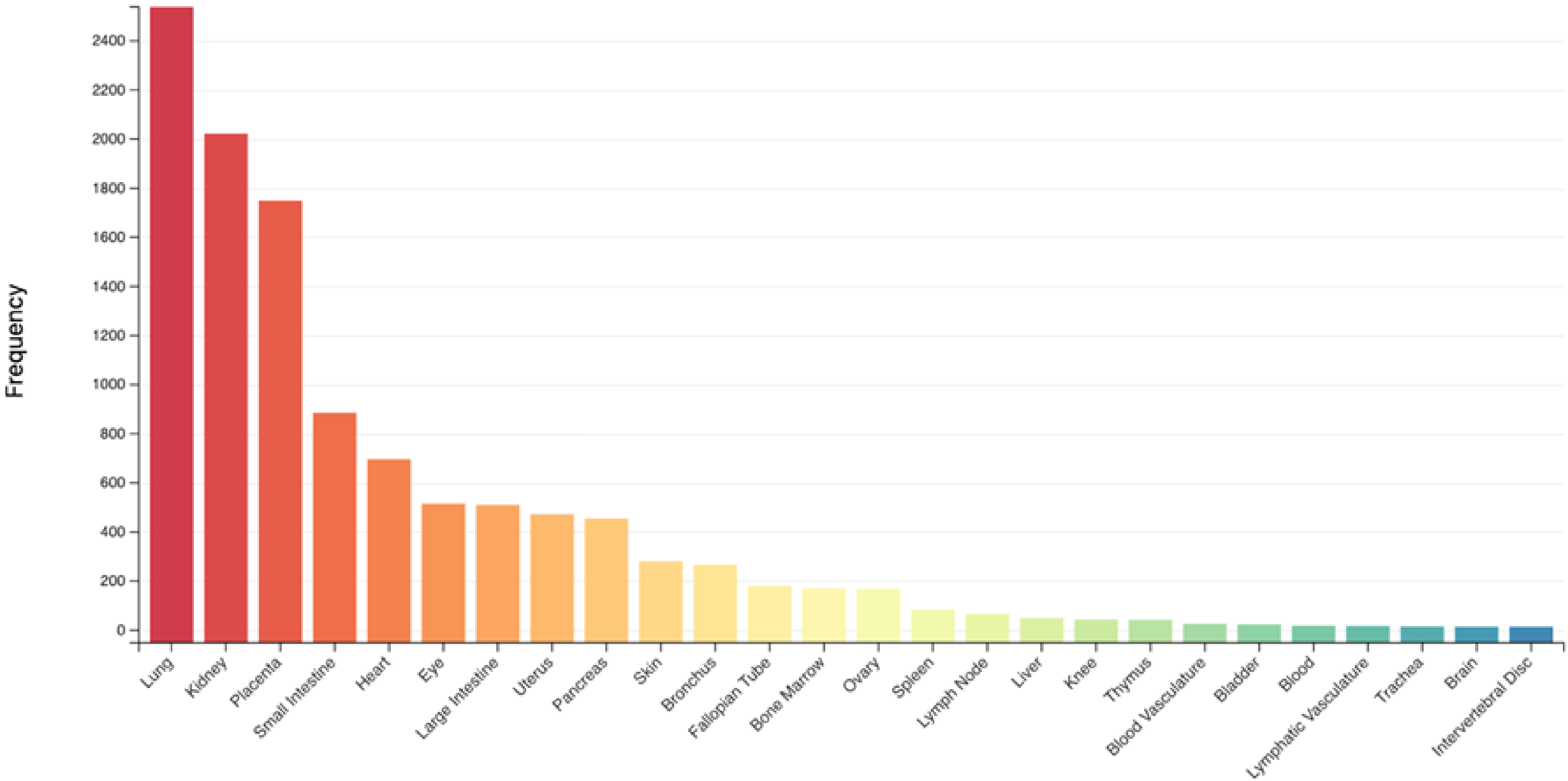


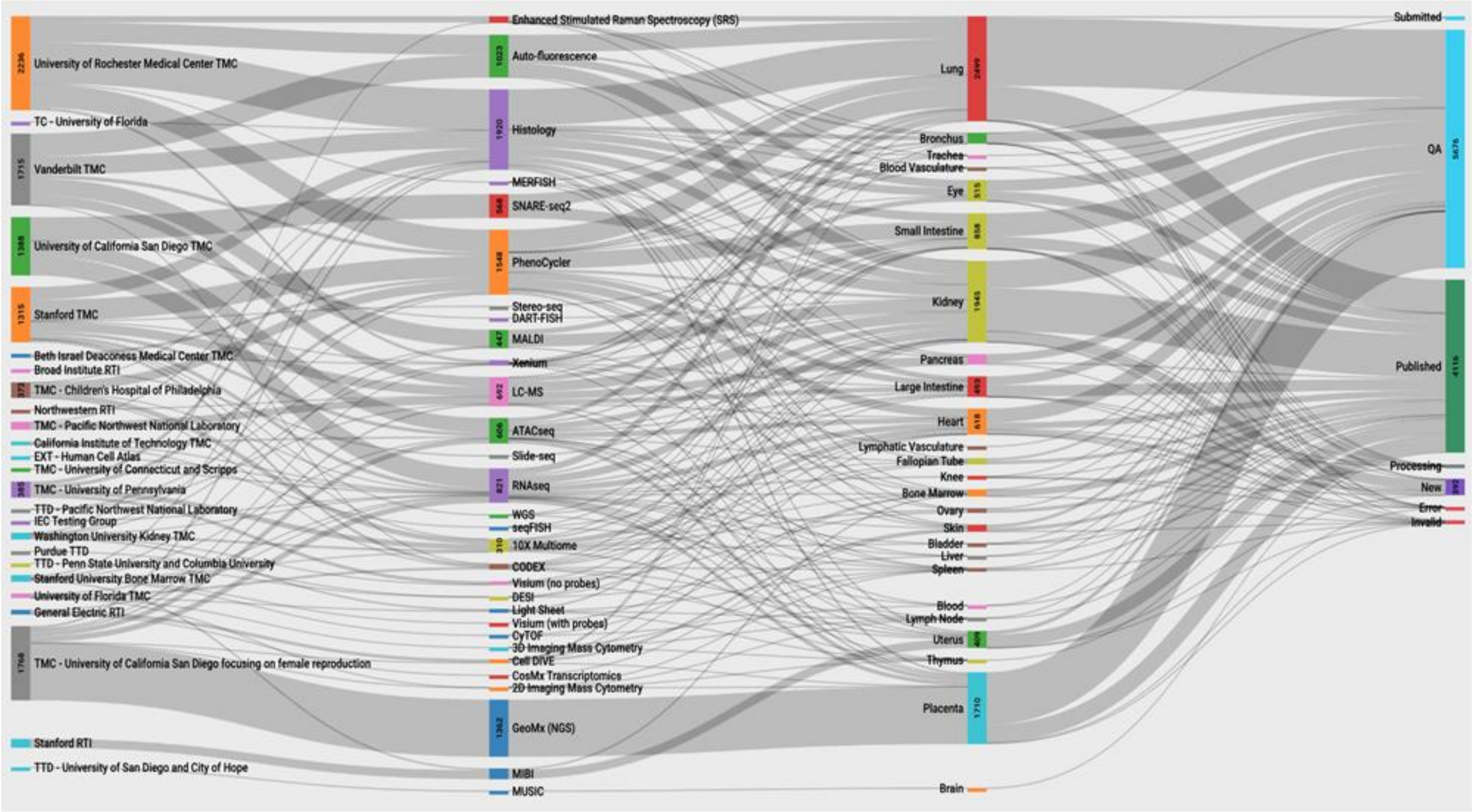
Data Types

24 currently
on the portal



Current Snapshot of Published/Soon-to-be Published HuBMAP Data




University of Rochester Medical Center TMC
2236
TC - University of Florida
Vanderbilt TMC
1715
University of California San Diego TMC
1388
Stanford TMC
1315
Beth Israel Deaconess Medical Center TMC
Broad Institute RTI
TMC - Children's Hospital of Philadelphia
Northwestern RTI
TMC - Pacific Northwest National Laboratory
California Institute of Technology TMC
EXT - Human Cell Atlas
TMC - University of Connecticut and Scripps
TMC - University of Pennsylvania
TTD - Pacific Northwest National Laboratory
IEC Testing Group
Washington University Kidney TMC
Purdue TTD
TTD - Penn State University and Columbia University
Stanford University Bone Marrow TMC
University of Florida TMC
General Electric RTI
TMC - University of California San Diego focusing on female reproduction
1768
Stanford RTI
TTD - University of San Diego and City of Hope
Enhanced Stimulated Raman Spectroscopy (SRS)
Auto-fluorescence
1023
Histology
1920
MERFISH
SNARE-seq2
PhenoCycler
1548
Stereo-seq
DART-FISH
MALDI
447
Xenium
LC-MS
692
ATACseq
606
Slide-seq
RNAseq
821
WGS
seqFISH
10X Multiome
310
CODEX
Visium (no probes)
DESI
Light Sheet
Visium (with probes)
CyTOF
3D Imaging Mass Cytometry
Cell DIVE
CosMx Transcriptomics
2D Imaging Mass Cytometry
GeoMx (NGS)
1362
MIBI
MUSIC
Lung
2499
Bronchus
Trachea
Blood Vasculature
Eye
515
Small Intestine
858
Kidney
1945
Pancreas
Large Intestine
Heart
618
Lymphatic Vasculature
Fallopian Tube
Knee
Bone Marrow
Ovary
Skin
Bladder
Liver
Spleen
Blood
Lymph Node
Uterus
Thymus
Placenta
1710
Brain
Submitted
QA
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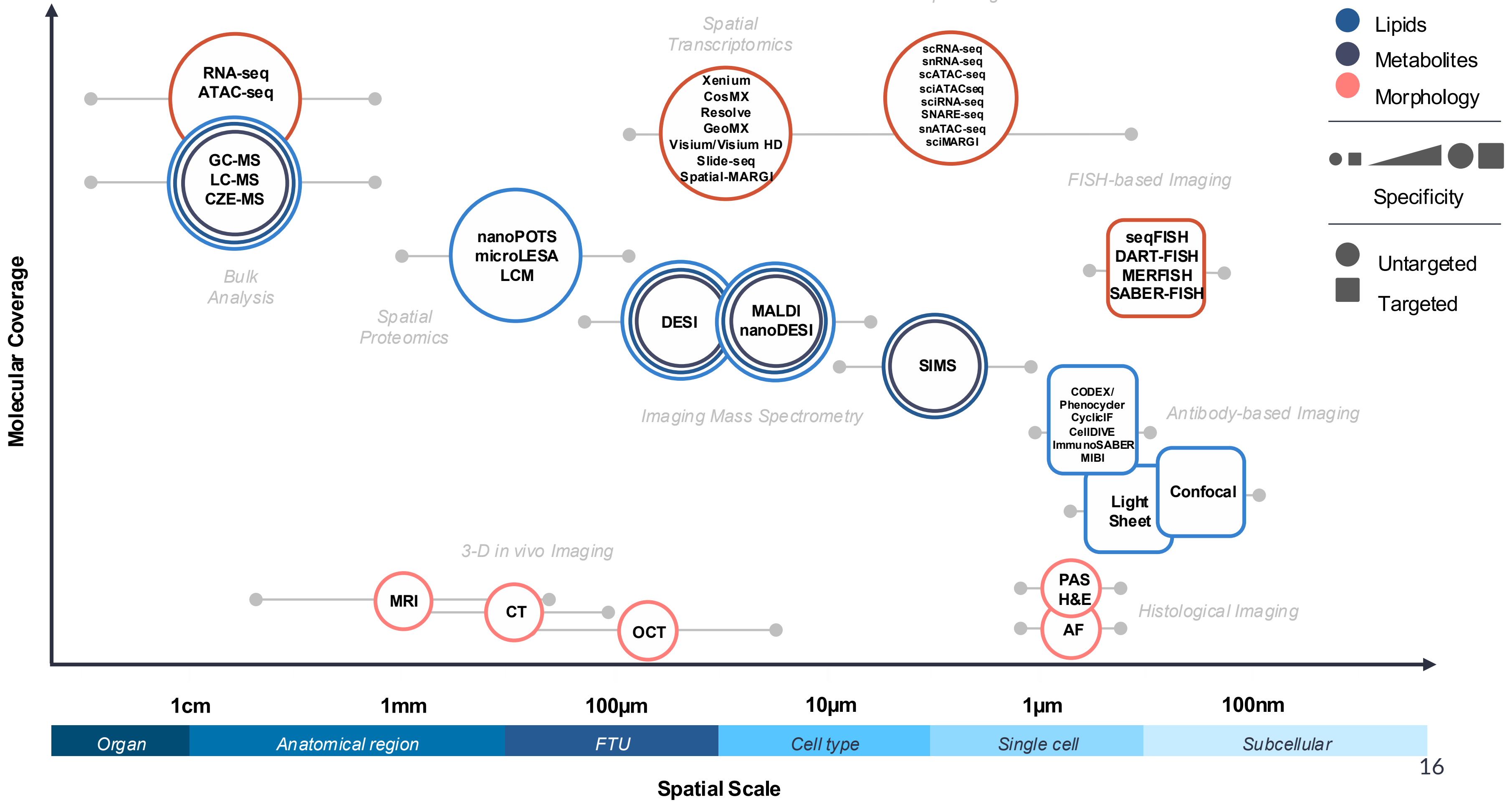
Published
4116
Processing
New
Error
Invalid

HuBMAP Technologies

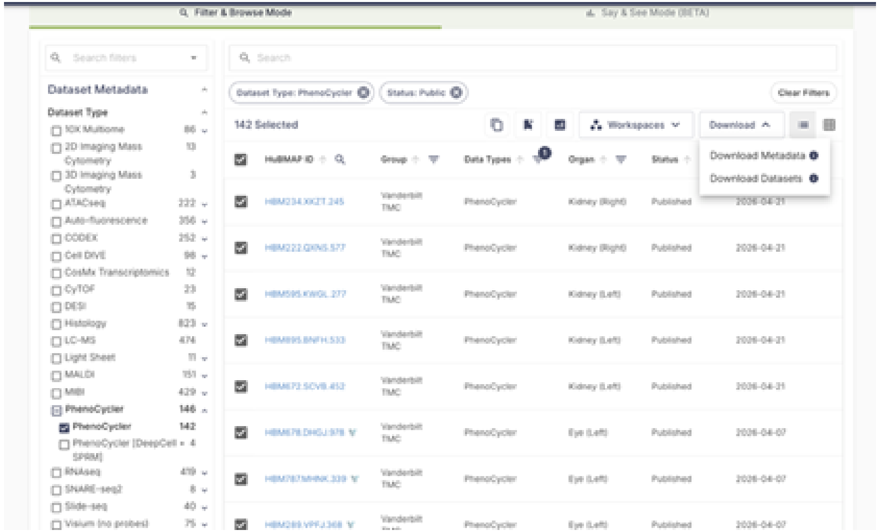


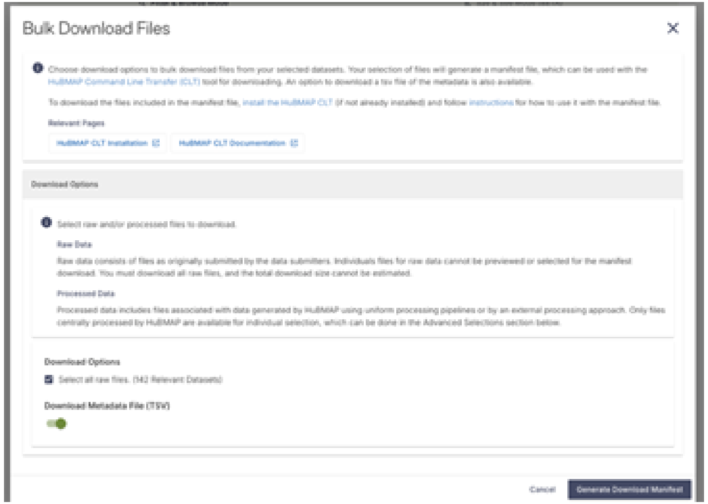
Examples of HuBMAP Datasets (Live Demos)

- Faceted search
- Example 1: Visium (no probes) - spatial transcriptomics (where RNA expression occurs in tissue)
  - Dataset HBM588.GLJR.289
- Example 2: PhenoCycler - spatial proteomics (where protein markers occur in tissue)
  - Dataset HBM849.GZFL.769
- Items of note:
  - Metadata
  - Provenance
  - Visualizations
  - Files/data products/data access - dbGaP access for sequencing data
  - Protocols and workflow details

Reusing HuBMAP Data

- Downloading HuBMAP data
  - Per the previous examples, you can use Globus to download non-sequencing data and dbGaP for raw datasets containing human sequencing data
- Bulk downloading HuBMAP data
  - HuBMAP offers a Command Line Transfer (CLT) utility for downloading multiple datasets at a time
    - This process uses a manifest.txt file generated from a faceted dataset search on the portal
    - Installing the CLT
    - Using the CLT





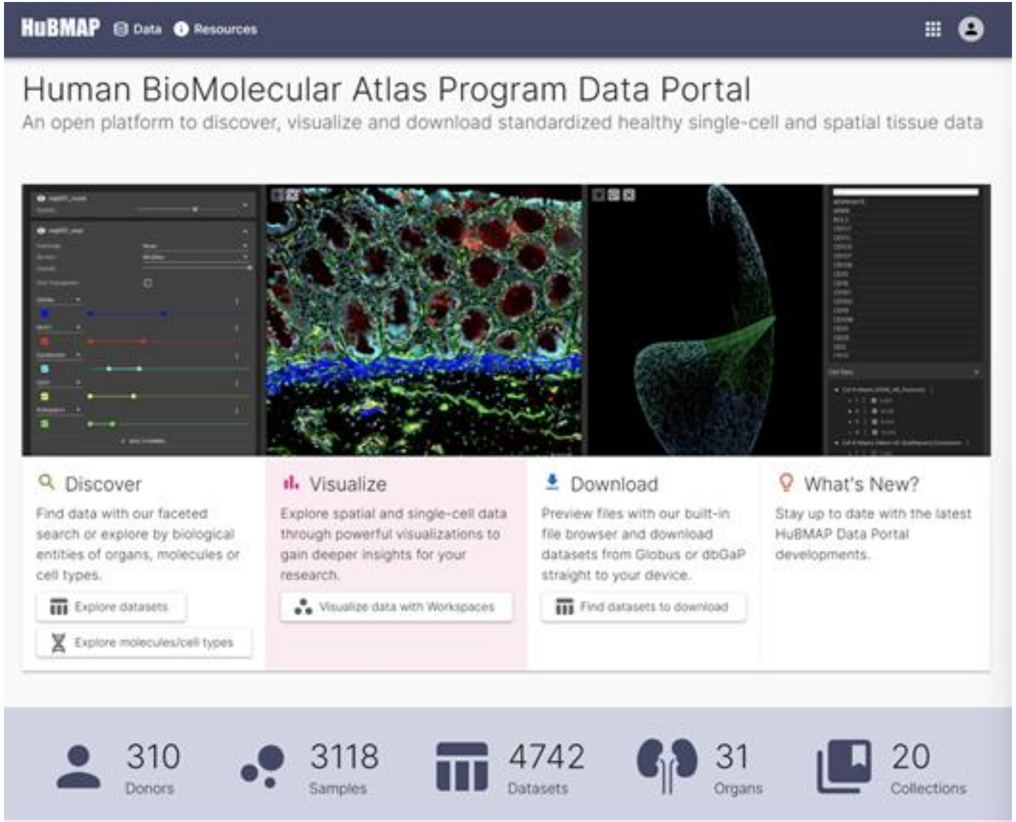
Come Visit Us!

HuBMAP Core Resources

- HuBMAP Data Portal
- HuBMAP Dataset Search
- HuBMAP Publications
- HuBMAP Documentation

Visualization and Analysis

- Vitessce
- HuBMAP Workspaces
- Azimuth for HuBMAP
- Human Reference Atlas



For any questions, email help@hubmapconsortium.org